DS	**Session**	Queue	**Question Information**	Level 1 Challenge 52

Question description

There is a bit string consisting of n bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le m \le 2 \cdot 10^5$
- $1 \le x_i \le n$

Input Format

The first input line has a bit string consisting of n bits. The bits are numbered $1,2,\ldots,n$.
The next line contains an integer m: the number of changes.
The last line contains m integers $x_1,x_2,\ldots,x_m$ describing the changes.

Output Format

After each change, print the length of the longest substring whose each bit is the same.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
001011
3
3 2 5
```

EXPECTED OUTPUT

```
4 2 3
```

Test Case 2

INPUT (STDIN)

```
101010
4
1 3 2 6
```

EXPECTED OUTPUT

```
2 4 2 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void pull(int k,int
l,int r)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
550
```

Test Case 3

NLOC

```
59
```

Code

solution.c

```c
#include <stdio.h>
#include <string.h>

void pull(int k, int l, int r) {}

int getMaxRun(char s[], int n) {
    int maxLen = 0, curLen = 1;
    for (int i = 1; i < n; i++) {
        if (s[i] == s[i - 1]) curLen++;
        else {
            if (curLen > maxLen) maxLen = curLen;
            curLen = 1;
        }
    }
    if (curLen > maxLen) maxLen = curLen;
    return maxLen;
}

int main() {
    char s[200005];
    if (scanf("%s", s) != 1) return 0;
    int n = strlen(s);
    int m;
    if (scanf("%d", &m) != 1) return 0;
    for (int i = 0; i < m; i++) {
        int x;
        scanf("%d", &x);
        x--;
        s[x] = (s[x] == '0') ? '1' : '0';
        int ans = getMaxRun(s, n);
        printf("%d%s", ans, (i == m - 1) ? "" : " ");
    }
    printf("\n");
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell

PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
001011
3
3 2 5
4 2 3
PS C:\C Programming>
```